

Memo 08: FVCOM-MP Effective Floc Size Coupling Using the ORIG_SED Floc Settling Treatment

FVCOM-MP WaterPACT Investigation

May 2026

Purpose

This memo records the technical implementation of a one-way FVCOM-MP coupling to the original FVCOM sediment flocculation treatment. The sediment module remains unchanged. The plastic module reconstructs the same local floc settling signal used by `mod_sed.F`, converts that settling speed to a Stokes-equivalent floc diameter, and uses the result only inside the MP-floc interaction pathway.

Theory Used

ORIG_SED floc settling speed

For cohesive sediment, `mod_sed.F` computes a local floc-enhanced settling velocity in `Calc_Wset_Floc`. The implemented empirical form is

$$W_s = W_0 F, \quad (1)$$

where

$$F = \frac{0.274 C^{0.48} S^{0.03} G^{0.22}}{(D_{50,\text{mm}})^{0.58}}. \quad (2)$$

Here C is suspended sediment concentration in kg m^{-3} , S is salinity, G is the turbulence/shear intensity argument described below, and $D_{50,\text{mm}}$ is the primary cohesive sediment diameter in mm. The existing ORIG_SED guards are preserved:

- if $F < 1$, set $F = 1$;
- if $S < 5$ or $S > 15$, set $F = 1$;
- cap the final settling speed at $W_s \leq 0.5 \times 10^{-3} \text{ m s}^{-1}$.

G computation

The MP implementation reconstructs the same G expression used in `mod_sed.F`. Element velocities are first mapped to nodes with `E2N3D(u,unode)` and `E2N3D(v,vnode)`. For each node and sigma layer,

$$j_e = \frac{m_f^2(u_n^2 + v_n^2)}{\max(d, 5)^{4/3}}, \quad (3)$$

with $m_f = 0.012$ and water depth d in m. The floc formula argument is then

$$G = \left(\frac{g \sqrt{u_n^2 + v_n^2} j_e}{V_{\text{cons}}} \right)^{1/2}, \quad (4)$$

with $V_{\text{cons}} = 10^{-6}$ and gravitational acceleration from the FVCOM control module. This is the same local calculation currently embedded in the sediment deposition routine, including the 5 m shallow-depth floor.

Stokes-equivalent floc size

The sediment model does not prognose floc diameter. The new MP option therefore uses an equivalent diameter obtained by assuming a spherical floc of constant effective density:

$$d_{\text{floc,eff}} = \left[\frac{18\mu W_s}{g(\rho_{\text{floc,eff}} - \rho_w)} \right]^{1/2}. \quad (5)$$

The parameter `PLAST_FLOC_RHO_EFF` supplies $\rho_{\text{floc,eff}}$. The default value is 1300 kg m^{-3} , matching the legacy MP aggregation kernel's floc density. The water dynamic viscosity is computed from the same bounded temperature-dependent kinematic viscosity already used in the MP aggregation kernel:

$$\nu = 10^{-6} \exp[-0.025(T - 20)], \quad 0.5 \times 10^{-6} \leq \nu \leq 2.0 \times 10^{-6}, \quad (6)$$

and $\mu = \rho_w \nu$. The implementation prevents the equivalent diameter from falling below the primary sediment D_{50} .

Code Change Summary

- Added optional plastic inputs `PLAST_FLOC_EFFECTIVE` and `PLAST_FLOC_RHO_EFF`. Missing keys preserve the old behavior: the effective floc-size path is off and the density defaults to 1300 kg m^{-3} .
- Added validation that `PLAST_FLOC_EFFECTIVE=T` requires `PLAST_FLOC=T` and `-DORIG_SED`.
- Added `Calc_ORIG_Floc_Effective_Cell` in `mod_plast.F` to reconstruct `ORIG_SED W_s`, compute G , and convert W_s to $d_{\text{floc,eff}}$.
- Updated MP aggregation so size exclusion, collision radius, sediment particle mass, and differential-settling kernels can use the effective floc diameter and reconstructed floc settling speed.
- Updated MP disaggregation so the maximum active floc size can come from the effective floc diameter field when the new switch is active.
- Updated aggregated MP settling so `wset_floc` stores a sediment-concentration-weighted reconstructed floc settling speed. The existing NetCDF variable `settle_vel_floc_*` can therefore be used in post-processing to back out $d_{\text{floc,eff}}$ with the same Stokes formula.
- Created `FVCOM_MP_test_run/INPUT/generic_plastic.d.inp` and `FVCOM_MP_test_run/INPUT/generic_sediment.d.inp` as the switch-on case. The `_c` files remain the comparison case.

Implementation Plan

1. Push the updated `mod_plast.F` to the HPC source tree and compile with the existing FVCOM-MP build options plus `-DSEDIMENT`, `-DORIG_SED`, and `-DPLASTIC`. The new MP option has a runtime guard, so a switch-on input will stop early if `ORIG_SED` is absent.
2. Keep the `_c` inputs as the comparison case: `generic_plastic_c.inp` and `generic_sediment_c.inp`. This case keeps `PLAST_FLOC=T` but does not activate the effective floc-size option.
3. Use the `_d` inputs as the switch-on case: `generic_plastic_d.inp` sets `PLAST_FLOC_EFFECTIVE=T` and `PLAST_FLOC_RHO_EFF=1300.0`; the new `generic_sediment_d.inp` switches the single sediment class to `SED_TYPE=cohesive`, `SED_SD50=0.004` mm, and `SED_WSET=0.1` mm s⁻¹. The sediment name is kept as `coarse_sand_1` so existing river-forcing variable names remain compatible.
4. Create or copy a `RUN_d` namelist from `RUN_c` and point it to `generic_sediment_d.inp` and `generic_plastic_d.inp`. Keep the hydrodynamic, river, source, and reporting settings aligned with `RUN_c` unless a deliberate sensitivity change is being made.
5. After a successful compile, start with a short HPC run of case `_d`. Confirm that the input guards pass, the run reaches the first few reports, and the existing `settle_vel_floc.*` output is present for post-processing.
6. Use the `_c` and `_d` outputs as a paired comparison. The `_d` run is the `ORIG_SED` effective-floc coupling case, and the `_c` run remains the non-effective MP-floc comparison.

Post-processing Note

When `PLAST_FLOC_EFFECTIVE=T`, the output variable `settle_vel_floc_<plastic name>` stores the effective aggregated MP floc-settling speed used by the plastic module. A post-processing script can recover the model-used equivalent floc diameter by applying the same Stokes formula above with the output `ambient_rho`, the configured `PLAST_FLOC_RHO_EFF`, and the same temperature-dependent viscosity closure. For multiple active sediment classes, this output represents the sediment-concentration-weighted effective settling speed used for aggregated MP settling, not a separate diameter for each sediment class.